

Lecture 18:
Policy Lab
Does rural electrification work? I

PPHA 34600

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Guest star: Kara Ross Camarena

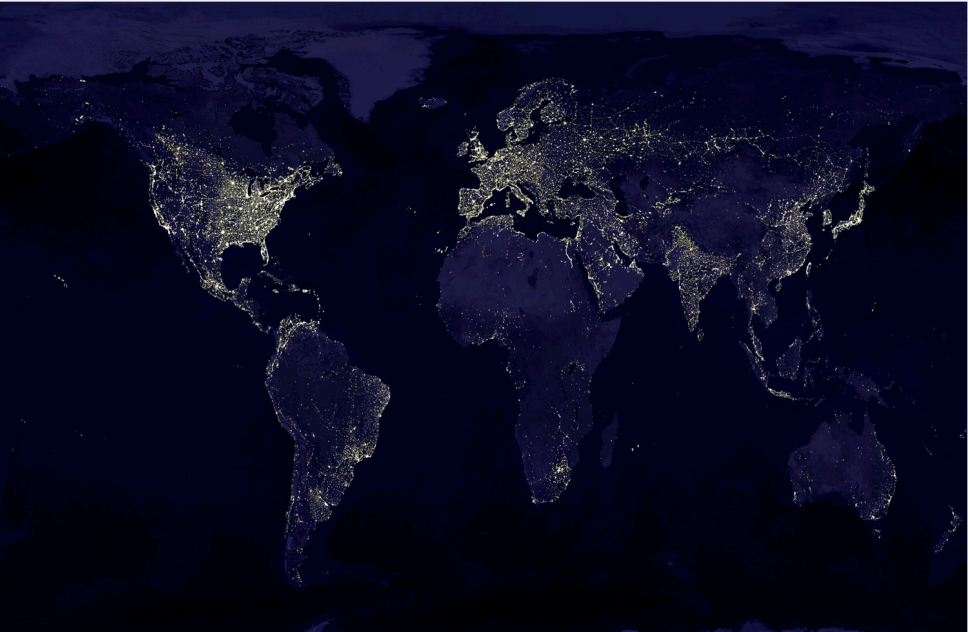
Harris School of Public Policy
University of Chicago

From last time: Machine learning for causal inference

We looked at several ways to incorporate ML into CI

- ① Generating (big) data
- ② Exploring heterogeneity
- ③ Improving research designs
 - ML works with SOO to handle functional form
 - And with SOU to aid in generating counterfactuals

The world at night



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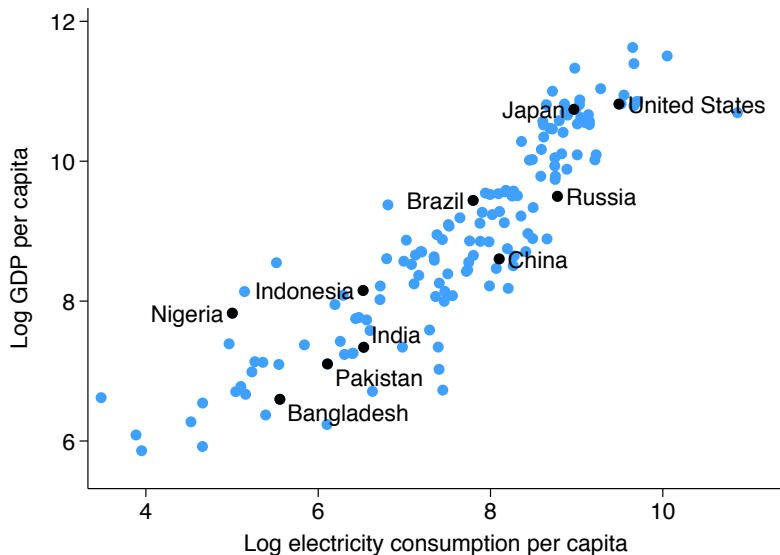
“Access to energy is essential to reduce poverty.”

— *The World Bank*

“By developing infrastructure that provides sustainable, reliable and affordable access to modern energy services, people, communities and countries can significantly improve their living standards and economic status.”

— *United Nations Development Programme*

Wealthy countries use more electricity per capita



Source: World Bank tables

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 - More (less) politically connected
 - Have other infrastructure (roads, etc)
 - Be wealthier (less wealthy)
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→ There are many forms of selection bias!

The ideal experiment

What would we do if we could do anything?

- Some kind of random assignment to electricity
- ... but even this is not straightforward

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 - Do we think about electrification in one location at a time?
 - Or do we consider general equilibrium?
- Actually ideal experiment probably requires multiple Earths
- Or at least a really large sample!

If you thought the *ideal* experiment was tricky...

An additional practical wrinkle:

- Randomizing electricity access is impractical!
 - Like roads, electricity works on a network
 - Putting a random segment in the middle of nowhere would not work
 - Simple randomization is therefore not really going to work
 - Not to mention that it's going to be extremely hard to randomize at any meaningful scale

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→ A quasi-experimental approach may be useful here

First paper: Dinkelman (2011)

This is the most prominent early econ paper estimating effects of RE:

- **Research question:** What is the effect of electrification on “the ability of the poor to use their labor resources for market production?”

→ AKA, what is the effect of RE on employment?

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 - Or more jobs in the household
 - Both in- and out-migration effects

Dinkelman (2011): Context

Dinkelman studies immediately-post-Apartheid South Africa:

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 - **2001:** 1/4 of households were newly connected to the grid
 - 2x as many households electrified as the first five years of the REA (US)
- Massive push towards rural electrification
- Opportunity for a natural experiment!

What did the electrification program look like?

- National Electrification Programme (NEP): electrify 300k households
- Implemented through Eskom, the monopoly utility
- Cost \$1.4bn; connected 470,000 hhs in KZN (1993–2003)
- Connections powerful enough to run a few (small) appliances
- Places that were more expensive to electrify got power later

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- A big switch from no power to power
- Very poor, marginalized groups getting access
- At the same time, many people getting access
- Important evidence from Sub-Saharan Africa

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- A big switch from no power to power
 - Very poor, marginalized groups getting access
 - At the same time, many people getting access
 - Important evidence from Sub-Saharan Africa
 - Also a unique setting
 - Nothing has ever really looked just like this context
- You should ask about external validity